


Workshop 1.

1. it's obvious that x & $\frac{1}{x}$ share same sign.

$$\text{thus } \left| x + \frac{1}{x} \right| = |x| + \frac{1}{|x|} \geq 2$$

$$2. \quad \sqrt{a+1} - \sqrt{a} = \frac{1}{\sqrt{a+1} + \sqrt{a}} < \frac{1}{2\sqrt{a}}$$

$$2\sqrt{a+1} - 2\sqrt{a} = \frac{1}{(\sqrt{a+1})^2 + (\sqrt{a(a+1)}) + (\sqrt{a})^2} < \frac{1}{2\sqrt{a}}$$

$$3. (a) \quad \phi = 1 + \frac{1}{\phi}$$

$$\phi^2 - \phi - 1 = 0$$

$$\phi = \frac{1 \pm \sqrt{5}}{2} \quad \text{with } \phi > 0$$

$$\phi = \frac{1}{2} (1 + \sqrt{5})$$

$$(b) \quad \frac{1}{x\phi} (\phi - x) = \frac{1}{x} - \frac{1}{\phi} = \frac{1}{x} + 1 - \phi \\ = f(x) - \phi$$

$$(c) \quad |f(x) - \phi| = \frac{1}{x\phi} |x - \phi| \\ \leq \frac{1}{\phi} |x - \phi|$$

$$(d) \quad |b_{n+1} - \phi| \leq \frac{1}{\phi} |b_n - \phi| \\ \leq \frac{1}{\phi^n} |b_1 - \phi| \rightarrow 0 \quad n \rightarrow +\infty$$

$$(e) \quad \frac{a_n}{a_{n-1}} = 1 + \frac{a_{n-2}}{a_{n-1}} \quad \text{set} \quad \frac{a_{n+1}}{a_n} = C_n$$

$$C_{n-1} = 1 + \frac{1}{C_{n-2}}$$

$$C_{n+1} = f(C_{n-1})$$

$$C_n \rightarrow \phi \quad \text{as} \quad n \rightarrow +\infty$$

$$4. \quad \lim_{n \rightarrow \infty} \left(1 + \frac{a}{n}\right)^n = \lim_{\frac{n}{a} \rightarrow +\infty} \left[\left(1 + \frac{a}{n}\right)^{\frac{n}{a}}\right]^a = e^a$$

$$5. \quad \forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N \quad |a_n - a| < \varepsilon$$

$$\Leftrightarrow \forall M < 0 \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N \quad |a_n - a| < M$$

$$\Leftrightarrow \forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N \quad |a_n - a| < M$$

$$6. (a) \quad \lim_{n \rightarrow +\infty} \sum_{k=1}^n \frac{1}{\sqrt{n^2 + k}} = \lim_{n \rightarrow +\infty} \frac{1}{n} \sum_{k=1}^n \frac{1}{\sqrt{1 + \frac{k}{n^2}}} \\ = \int_0^1 \frac{1}{1} = 1$$

$$(b) \quad \lim_{n \rightarrow +\infty} \sum_{k=1}^n \frac{1}{(n+k)^2} = 0$$

Workshop 2.

1. a_n is bounded above. thus $a = \sup A$ exists.
 $\forall \varepsilon > 0: \exists N_\varepsilon$ s.t. $a - \varepsilon < a_{N_\varepsilon} < a$

since a_n monotone increase

$\forall \varepsilon > 0: \exists N_\varepsilon$ s.t. $\forall n \geq N_\varepsilon, |a - a_n| < \varepsilon$

2.

Modk 1.

$$\sum_{k=1}^{\infty} a_k x^k$$

$$a_k = \begin{cases} 0 & k \neq n^3 \\ 3^k & k = n^3 \end{cases}$$

$$\lim_{n \rightarrow \infty} n(x^n - x^{2n})$$